

Index Futures Trading Restrictions and Spot Market Quality: Evidence from the Recent Chinese Stock Market Crash

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Abstract

Using a difference-in-difference approach, we find that restrictions placed on CSI 300 and CSI 500 index futures trading during the recent Chinese stock market crisis deteriorated spot market quality. This is more so for the September trade restriction. Our results can be explained by the sudden risk exposure faced by alpha-strategy traders. They stop trading spots after the index futures trading restrictions, hence worsening the spot market quality.

Keywords: Index futures; liquidity; market quality; difference-in-difference

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1 Introduction

In a frictionless market, derivatives trading that provides exposure to the same source of risk as the underlying asset will not affect the underlying market. When frictions do exist, however, the trading of even a simple derivatives such as index futures may affect the underlying asset's market quality. On the positive side, Silber (1985) argues that index futures trading may induce a low cost for market makers to hedge their inventory risk, hence narrowing the spread in the spot market. Grossman and Miller (1988) and Grossman (1989) state that index arbitrage between the spot and futures markets may improve spot market depth. Froot and Perold (1995) build an information model to show that market depth can be increased through a more rapid dissemination of market-wide information due to the introduction of index futures. More recently, Banerjee and Graveline (2014) show theoretically that when an underlying asset is scarce, derivatives trading can reduce the underlying asset scarcity and alleviate the spot price distortion. On the negative side, according to informed trader theory (Gorton and Pennacchi (1993); Subrahmanyam (1991)), index futures markets provide market-wide risk exposure with a lower cost than spot markets, hence the introduction of a futures market may draw away uninformed traders with well-diversified portfolios from a spot market. This implies that, as a result, the proportion of informed traders in the spot market increases which causes market makers to face a higher adverse selection cost and reduces liquidity in the spot market after the introduction of index futures. Clearly, the effect of index futures trading on spot market quality warrants empirical tests as theories have different perspectives and diverge on this important issue.

Empirical evidence is, however, rather limited and inconclusive. Bessembinder and Seguin (1992) examine the effect of index futures trading activities on spot market volatility and attribute the reduction of volatility to increased liquidity from futures trading. Roll

et al. (2007) find that spot market liquidity and the futures-spot basis can Granger-cause each other. In contrast, Jegadeesh and Subrahmanyam (1993) test the direct effect of index futures trading on average bid-ask spreads. They find a 3.7% increase of spread on the introduction of index futures trading. However, contrary to informed trading theory, they fail to find a significant increase in the adverse selection component of the spread. Contradicting results are also found in the voluminous literature on the relationship between index futures trading and spot market volatility (see Chen et al. (2013) for the latest review). One important reason for these empirical inconsistencies could be the existence of confounding factors that may also affect the spot market quality along with the futures trading introduction. As such, conventional event study methods such as using dummy variables in a regression model may not be suitable for such analyses. Another important reason is that studies using index futures introduction as an event are joint tests of futures market efficiency and the role of futures trading on spot market quality. As shown in Figlewski (1984), disequilibrium price discounts exist in early index futures trading, possibly due to “unfamiliarity with the new markets and institutional inertia in developing systems ...”. Yang et al. (2012) also show that, at the early stage of Chinese CSI 300 index futures trading after introduction, the futures market does not function well in terms of information transmission and price discovery.

This article provides additional empirical evidence on the effect of index futures trading on spot market quality using the 2015 summer Chinese stock market crash as a quasi-natural experiment. Critics of the role of index futures in the crash argue that the index futures market serves as a venue of speculative trading (short selling) and that huge profits are made by few market “manipulators” through simultaneously dumping stocks in the spot market, accelerating the spot market crash. From late August 2015, under intensified social pressure, the China Financial Futures Exchange (CFFEX) adopted a series of harsh restrictions on the scale of non-hedging open positions and effectively terminated index

futures trading in China. We see this event as a unique opportunity to re-examine the effect of index futures trading on spot market quality because previous literature mainly examines the issue by looking at the introduction, not the termination, of index futures trading. As the futures market shifts from a well-functioning state to a nearly full stop, our study avoids the joint test problem discussed earlier.

To eliminate the effect of confounding factors and identify the role of futures trading on market quality, we adopt a difference-in-difference (DID) approach using index component stocks as the treatment group and non-component stocks as the control group. This is the same approach as Harris (1989), Laatsch (1991), Kumar et al. (1995), and more recently Xie and Mo (2014). Note that a critical assumption underlying the DID approach in our context is that investors can not cross-hedge, i.e. restrictions on index futures trading should not affect the trading of non-component stocks. If this assumption does not hold, then estimates based on the DID approach could be misleading. Fortunately, there are at least two reasons why this assumption holds in our case. First, on April 16th, 2015, the CFFEX launched two new index futures, the IC and IH futures corresponding to the CSI 500 and Shanghai 50 indices receptively. Although the Shanghai 50 index is a subset of the CSI 300 index which covers large and median-size firms, the CSI 500 index is composed of small-size firms. Together the CSI 300 and CSI 500 index cover more than 90% of the total market value in the Chinese A share market. In our sample period, all these three index futures have been traded, hence the likelihood of cross-hedge is rather small. Second, since July 6th, early period in our sample, the CFFEX has restricted that the value of an investor's sell-to-hedge portfolios not exceed the value of the investor's holding of the corresponding index component stocks. This unusual restriction effectively rules out cross-hedge in our sample period, making the DID approach very suitable for the question under study.

Using high frequency data, we construct a series of liquidity and efficiency indicators to

measure market quality. Statistical results show that, controlling for other variables, after the effective termination of the Chinese index futures trading, the index component stock market quality is significantly reduced compared with the non-component stock market quality. We then discuss possible explanations. We reject the informed trader theory and index-arbitrage theory. Then a new explanation is offered. When index futures trading is restricted, investors who previously trade on alpha strategies, i.e., long a stock portfolio and short index futures, are exposed to huge systematic risk and hence will stop trading spots, causing the spot market quality to be worse.

Our study complements the empirical literature of the effect of derivatives introduction on spot market. Conrad (1989) finds a price increase and volatility decrease after option introduction. Figlewski and Webb (1993) find that options introduction reduces short sale constraints of the spot and increases transaction and information efficiency. Kumar et al. (1998) and De Jong et al. (2006) also find that options trading improves spot market information efficiency and market quality. However, Bollen (1998) didn't find a significant effect of option introduction on spot return variance. Boehmer et al. (2015) document a decrease of spot market quality after introduction of single name credit default swaps. It is clear that after decades of derivatives trading, its impact on the underlying market quality is still a debatable question. Our study adds to the literature by investigating the role of derivatives termination, rather than introduction, on spot market quality.

Our study also complements the literature of derivatives regulations during a market crisis. Kleidon and Whaley (1992) argue that the delinkage of spot, futures and options markets is due to outdated market order processing, hence futures trading restrictions are not helpful for market integration in a crisis. Program trading involving portfolio insurance using index futures is also at the centre of debate after the US 1987 market crash. The SEC report (Securities and Commission (1988)) and the Brandy report (1998) criticize the portfolio insurance as deleterious to market quality. However, the CME report (Miller et al.

(1987)) and CFTC report refute these arguments. The extreme measures of futures trading restrictions by the Chinese regulators in the latest stock market crash and the heated debate before and after the adoption of these restrictive measures highlight the fact that the role of index futures trading on the spot market quality is not a resolved question and deserves further investigations.

The rest of the article is organized as follows. Section 2 introduces the background and the regulatory measures adopted to curb index futures trading. Section 3 describes our research methodology, including the stock matching method and the DID approach. Section 4 presents the major results and possible explanations are offered. Section 5 concludes.

2 The Recent Chinese Stock Market Crash

The Chinese stock market has recently taken a rollercoaster ride, as shown in Figure 1. From the beginning of 2015 until Mid-June, the CSI 300 Index, which represents the broad Chinese A share market, rose from 3500 to a high of 5178. Then it collapsed, losing over 34% in twenty days, with 1000 points erased in one week alone. On July 5, the Chinese government adopted a series of supportive measures only to see the market drop another 1000 points in the third week of August. Attempts were made to restrict high frequency program trading and to investigate abnormal individual trading account, but failed. During the crisis, almost half of the listed stocks lost more than 50% of their pre-crash market value, with the biggest loss more than 77%. From mid-June to mid-September, on average, one day in every four trading days saw more than one thousand stocks (about one third of the total number of stocks traded in the market) lose 10%, i.e., hit the lower bound of the daily price fluctuations. By any means, this crash is among the most dramatic stock market crises in history.

[Insert Figure 1]

Under heavy pressure from both regulators and the public, the CFFEX trading restriction announced on August 25 that from August 26, three measures would be adopted to curb speculative trading in the index futures market¹. First, the initial margin for non-hedging trades would be raised from first 10% to 12%, then 15% and finally 20% over the following three days. Second, any single day's total opening position greater than 600 contracts would be considered as abnormal trade and be subject to increased scrutiny. Third, the clearing fees for intraday trades would be adjusted upward to 1.15 basis points.

With the crisis developing further and the effect of government bail out measures quickly diminishing, September 2 saw the CFFEX announce yet another round of measures to curb speculative trade in the futures market. First, from September 7, any single day non-hedging trading of over 10 contracts would be considered abnormal. Second, the initial margins for all non-hedging trading would be raised from 30% to 40%, and the initial margins for hedging trades would be raised from 10% to 20%. Third, the clearing fees for intraday trades would be adjusted from 1.15 to 23 basis points. With these two rounds of measures, index futures trading in China nearly came to an almost complete stop, as shown in figure 2. This rare regulatory practice that put direct limits on the number of trades provides a unique setting to examine the impact of index futures trading in an order-driven spot market.

[Insert Figure 2]

¹China launched its CSI 300 index futures on April 16, 2010, and then introduced the CSI 500 and Shanghai 50 index futures five years later. Guo et al. (2013) show that the CSI 300 index futures has performed well in terms of its price discovery function relative to the spot market and the Singapore A50 futures market.

3 Data and Methodology

Our goal is to compare the market quality change of individual stocks within the index (component stocks) against that of a control sample of stocks that are not in the index (non-component stocks) before and after the two rounds of CFFEX announcements. Since the CSI 500 and Shanghai 50 index futures were just launched on April 16, 2015, we omit the first two weeks of trade to avoid possible initial trading inefficiencies as documented in Figlewski and Webb (1993) and Yang et al. (2012). Thus, our whole sample spans 05/01-09/30. We then divide the sample period into three sub-periods based on the CFFEX policy announcement dates: 05/01-08/25, 08/26-09/06, and 09/07-09/30. Because almost all component stocks of the Shanghai 50 index are also included in the CSI 300 index and the two indices are highly correlated, we believe that the termination of trading of either the CSI 300 index or the Shanghai 50 index futures would generate a similar impact on spot market quality. Hence for brevity we do not report the effect of the Shanghai 50 index futures trading in separately.

Since all sub-periods are rather short, we use intra-day data for the empirical analysis. The 3-second (for stocks traded in the Shenzhen Stock Exchange) and 5-second (for stocks traded in the Shanghai Stock Exchange) quoted stock price and trade volume data and all firm information are from CSMAR, a leading financial data provider in China. Stock index adjustment information is from WIND, another major Chinese financial data provider. We apply the following filters to the raw data: financial stocks, stocks with at least five consecutive non-trading days, stocks under special treatment, and stocks lacking data required for the following regressions are deleted. Finally, any stocks added to or deleted from the three indices during the sample period are not included. After filtering we are left with 1037 stocks in the sample.

We use six measures, namely market tightness (bid-ask spread), market breadth (volume

and turnover), market depth (depth and the Amihud illiquidity ratio² (Amihud (2002)), denoted by *Amihud*), and market efficiency (the high frequency volatility), to measure daily stock market quality.

Specifically, the market tightness index is constructed using the relative bid-ask spread(RSP), calculated as

$$RSP_{i,t} = \frac{1}{N} \sum_{n=1}^N \frac{Ask_{i,n} - Bid_{i,n}}{(Bid_{i,n} + Ask_{i,n})/2}$$

where $Ask_{i,n}$ and $Bid_{i,n}$ are the ask and bid prices of stock i at intra-day time n . N is the total times of quotations at trading date t .

Market breadth is described by trade volume and stock turnover. *Turnover* of firm i at date t is calculated as

$$Turnover_{i,t} = \frac{TS_{i,t}}{CS_{i,t}}$$

where $TS_{i,t}$ is the number of traded shares of stock i at date t and $CS_{i,t}$ is the number of circulating shares of firm i at date t ³.

Market depth is computed using Amihud illiquidity ratio. Suppose that prices $p_{i,t,0}$, $p_{i,t,1}$, $p_{i,t,2} \cdots p_{i,t,n}$ are $n+1$ observed intra-day log prices of stock i at date t . Define the intra-day return as $r_{i,t,j} = p_{i,t,j} - p_{i,t,j-1}$. The Amihud illiquidity ratio is then given by

$$Amihud_{i,t} = \frac{1}{n} \sum_{j=1}^n \frac{|r_{i,t,j}|}{Volume_{i,t,j}}$$

where $Volume_{i,t,j}$ is the traded RMB volume of stock i from time $j-1$ to j at date t . The *Amihud* indicator is supposed to capture the price impact of trades. In a deep market with low *Amihud*, a large volume of trades generates limited price impact.

²The Amihud illiquidity ratio is a widely accepted measurement of the trading volume needed to move the stock price. See, for example, Goyenko et al. (2009) for stock liquidity and Marshall et al. (2011) for commodity liquidity. In both studies, Amihud liquidity is shown to perform better than other measures in estimating the price impact of trades.

³The Chinese stock market has some non-circulating stocks that are held by large shareholders and are non-tradable at the secondary market.

Two additional market depth indices of firm i at date t are calculated as:

$$Depth1_{i,t} = \frac{1}{n} \sum_{j=1}^n \frac{(S_{1,i,j,t} \times SV_{1,i,j,t} + B_{1,i,j,t} \times BV_{1,i,j,t})}{2}$$

$$Depth5_{i,t} = \frac{1}{n} \sum_{j=1}^n \frac{\sum_{m=1}^5 (S_{m,i,j,t} \times SV_{m,i,j,t} + B_{m,i,j,t} \times BV_{m,i,j,t})}{2}$$

where for each date t , the j is intra-day trading time of date t , $S_{m,i,j,t}$ is the m th ask price at time j of, $SV_{m,i,j,t}$ is the quoted volume at the m th ask price at time j , $B_{m,i,j,t}$ is the m th bid price at time j , $BV_{m,i,j,t}$ is the quoted volume at the m th bid price at time j . $Depth1$ and $Depth5$ use the optimal first and the optimal first five, respectively, bid and ask prices and volume for computation.

In the literature, e.g. Boehmer et al. (2015) and Foley and Putnins (2016), market efficiency is captured by three indices including high frequency volatility (HFV), variance ratio and absolute autocorrelation of intra-day returns. We decide not to use the latter two measures because they try to capture the return deviation from a random walk. However, during a market crash as in our study, stock price movement can not follow a random walk. Hence, we choose HFV to measure market efficiency as it is a proxy for noise and temporary deviations of prices from their equilibrium value due to trading frictions. Following the literature, we calculate the HFV as

$$HFV_{i,j,t} = \log(midpoint_{i,j,t}) - \log(midpoint_{i,j-1,t})$$

where the *midpoints* are the mid prices between 10-second bid and ask prices.⁴

Throughout our empirical analysis, realized volatility (RV) and firm size ($Fsize$) are used as control variables since previous literature has shown that these variables are highly correlated with stock liquidity. Following Andersen and Bollerslev (2003), we use 1-minute

⁴In case quotes at the tenth second are missing, we use the previous midpoint and lag midpoint prices to smooth fit the missing price. For example, if a stock has quotes at the 9th second and the 12th second, we then calculate the midpoint price at the 10th second as $mid_{10s} = mid_{9s} + \frac{mid_{12s} - mid_{9s}}{12-9}$.

intra-day log returns of stock i to calculate the realized volatility of stock i at date t :

$$RV_{i,t} = \sqrt{\sum_{j=1}^n r_{i,t,j}^2}.$$

Because the tick size is fixed as 1 cent (RMB), some liquidity measures may also be related to the nominal price, so we control for the inverse of price of stock i at date t , denoted by $Invp_{i,t}$. We also control for the return of stock i at date t , denoted by $return_{i,t}$.

We then use stock information in the pre-restriction sub-period of 05/01-08/25 to match stocks. Our matching method follows Xie and Mo (2014). This matching procedure guarantees that each component stock is matched with one non-component stock. Note that because the matching is implemented independently for the CSI 300 index and the CSI 500 index, it is possible that some non-component stocks are matched with both a CSI 300 component stock and a CSI 500 component stock. Specifically, the matching is composed of the following steps. First, we obtain estimator $\hat{\alpha}$ by running the regression below:

$$Y_i = \alpha_0 + \alpha \cdot X_i + \epsilon_i$$

where Y_i is the time series average of market quality measurement of stock i before August 26, 2015. X_i is a control variable vector with Each element being the time series average of the control variables for stock i before August 26, 2015. Note that, since Chinese financial institutions in the index tend to be much larger in firm size than other public firms, it is hard to match their stocks with non-component stocks. In the data filtering process, we have deleted all financial stocks.

We next calculate $Distance_{i,j} = (X_i - X_j)' \hat{\alpha}' \hat{\alpha} (X_i - X_j)$ for $i \neq j$ where stock i is a component stock of either the CSI 300 or the CSI 500 index, and stock j is a non-component stock. We restrict stock pairs (i, j) to being in the same industry as specified in the 2012 Chinese Securities Regulatory Commission (CSRC) class one security industry code (SIC). Matched pairs are then constructed by minimizing $Distance_{i,j}$. This procedure is repeated for both the CSI 300 index and the CSI 500 index until all component stocks are matched.

After the matching is complete, we are left with 161 pairs for the CSI 300 index and 262 pairs for the CSI 500 index. Table 1 presents the summary statistics of our matched samples during the sample period. Consistent with the notion that the CSI 300 index represents the largest firms in the Chinese A-share market, the CSI 300 component stocks tend to be more liquid, less volatile, and more efficient than CSI 500 component stocks, which are in turn more liquid, less volatile and more efficient than non-index-component stocks. However, matched stocks for both indices, in particular the CSI 500 index, exhibit roughly similar characteristics.

[Insert Table 1]

It is beneficial to visually examine the time series of these market quality indicators before a rigorous statistic test. Figure 4 and Figure 5 illustrate the time series of these quality indicators for matched pairs for the two indices, respectively. Two observations can be made: first, the CSI component stocks are more liquid and more efficient than non-component stocks, both before and after the market crash, consistent with component stocks being preferred assets for market investors, presumably index arbitragers and stock fund managers; and second, after the CFFEX adopted trade restriction regulations, the liquidity spread between component and non-component stocks narrows, and the efficiency gap between non-component and component stocks shrinks, suggesting that trade regulations may hurt spot market quality.

[Insert Figure 4]

[Insert Figure 5]

In the next section, a standard DID analysis is applied to the matched samples to identify the net effect of futures trading restriction on the spot liquidity. Specifically, for each event, we run the following DID regressions for the CSI 300 and CSI 500 indices:

$$\log(MKTQ_{i,t}) = \beta_0 + \gamma_i \Sigma_{i=1}^{N-1} F_i + \alpha_{T_i} \Sigma_{i=1}^{T-1} T_i + \beta_c X_{i,t} + \beta_i D_i + \beta_t D_t + \beta_{tr} D_i \times D_t + \epsilon_{i,t}$$

where $MKTQ_{i,t}$ represents a measurement of market quality. Individual stock dummies, F_i , are used to control stock fixed effect and time dummies T_i are used to control time fixed effect. $X_{i,t}$ is a control variable vector. $D_i = 1$ if i is a component stock of CSI 300, $D_i = 0$ otherwise. $D_t = 1$ if $t \geq \text{Event date}$, $D_t = 0$ otherwise. The coefficient β_{tr} captures the net effect of index futures trading restriction on spot liquidity. Bertrand et al. (2004) point out that ignoring the time series autocorrelation would underestimate the standard deviation of the coefficient of treatment effect, so we cluster the standard deviation by firm to deal with this problem.

4 Index futures Trading Restrictions and Spot Market Quality

4.1 The Difference-in-Difference Analysis

Table 2 reports the regression result for relative bid-ask spread. In all regressions, the coefficients of the control variables have the expected signs and are consistent with the literature: stocks with higher volatility, more trade volume and cheaper price are associated with wider bid-ask spreads. The coefficients of D_i are significantly negative, consistent with Figure 4 indicating that index component stocks have, in general, smaller spreads than non-component stocks. The coefficients of D_t are positive at the 1% significance level, suggesting that after the announcements of trading restrictions the market bid-ask spread widens. The coefficients of DID variable, $D_i \times D_t$, are positive and significant at the 10% level for the September restriction for the CSI 300 index, and are more significant for the CSI 500 index. These strongly suggest that futures trading restrictions reduce the liquidity of component stocks relative to that of non-component stocks. More interestingly, a comparison of the DID coefficients across the two events suggests that, the September

trading restrictions have a greater effect than the August restriction. This is mostly likely because the September round limits daily non-hedging opening positions to a mere 10 contracts, compared to the initial restriction to 600 contracts in August, and dramatically increases the initial margin and clearing fees. Hence, we interpret the stronger effect of the second event as further evidence that futures trading restrictions increase the spot spread. In addition, the DID coefficients in the CSI 500 index case are all larger than those in the CSI 500 index case, suggesting that the bid-ask spread effect of futures trading termination is greater for the newly launched CSI 500 index futures.

[Insert Table 2]

Table 3 and Table 4 report the results for market breadth. Consistent with the intuition, higher volatility, larger firm size and cheaper price are associated with higher trade volume. On average, component stocks have higher trade volume but lower turnover than non-component stocks. After futures trading restrictions, for both CSI 300 and CSI 500 indices, the trade volume and turnover decrease significantly. The DID coefficient, $D_i \times D_t$, are negative significant in almost all regressions, suggesting that the futures trading restrictions are associated with lighter trading and narrower market breadth. Different from the case of relative bid-ask spread, the relation between futures trading restriction and spot turnover is stronger for the CSI 300 case than the CSI 500 case.

[Insert Table 3]

[Insert Table 4]

Table 5 to Table 7 report the result for market depth measures. Higher volatility and smaller firm size are associated with lower market depth. After futures trading restrictions, market depth decrease significantly in general, as seen from the coefficient of D_t . The DID coefficients $D_i \times D_t$ suggest a very strong negative effect of futures trading restriction on the depth of component stocks relative to non-component stocks. For the CSI 300 index case, this negative association is stronger for the September restriction than the August

restriction.

[Insert Table 5]

[Insert Table 6]

[Insert Table 7]

Table 8 reports the results for market efficiency, namely the HFV. Larger stocks with cheaper price and higher trade amount are typically less efficient than otherwise. Except for the August restriction for the CSI 500 index, component stocks are more efficient than non-component stocks, and the futures trading restrictions deteriorate market efficiency in general. The DID coefficients, $D_i \times D_t$, suggest that index futures trading restrictions increase the high frequency volatility more for component stocks relative to non-component stocks. Again, this positive relation is stronger for the September restriction than for the CSI 300 index.

[Insert Table 8]

To summarize, the above results clearly suggest a deterioration of spot market quality after the two rounds of index futures trading restrictions. This finding is strong as we control for most known factors, and more importantly, the DID approach as applied to matched samples is able to get rid of potential confounding factors which are not well controlled for in previous studies.

4.2 Discussions

Why do restrictions on index futures trading deteriorate the spot market quality? First of all, the informed trader theory does not apply as it predicts the opposite from our findings. It is tempting to interpret our results in line with Grossman and Miller (1988) and Grossman (1989) who argue that the index arbitrage trading between spot and futures markets improve the liquidity of spot markets. Restrictions of index futures trading impose high cost to index arbitrage, thus reducing the spot quality. However, a careful thought

would suggest that index arbitrage trading cannot be the main reason for our findings because since July 2015, more than one month before the events we considered, stock short selling has effectively been frozen by regulators. With the futures basis kept at negative levels throughout the market crash⁵, this implies that index arbitrage trading, which requires shorting stocks in this case, has already become very thin before the CFFEX announced the two rounds of trade restrictions. Hence the liquidity impact difference between component stocks and non-component stocks cannot be explained by the index arbitrage story.

[Insert Figure 3]

Besides index arbitragers, hedgers are not willing to open new positions either as a deep negative basis guarantees big losses at the maturity. Those left in the market are primarily “alpha traders”. Indeed, our conversation with market professionals confirms that, before the market crash, many institutional investors adopt the alpha strategy, i.e. hold a “systematic-risk-neutral portfolio that long stock portfolios and short index futures to profit from the excess return of stock portfolios relative to the index. Because the index component stocks are more liquid and more correlated with the index futures than non-component stocks, the stock portfolios in the alpha strategy are mainly composed of index component stocks. Many alpha-strategy traders, mostly mutual funds and hedge funds, registered as “speculators” with CFFEX to avoid the complex administrative procedures involved in registration as “hedgers”. These alpha-traders should have been neutral from the crash risk if there is no restriction on the futures trading. But if the index futures trading is restricted and this restriction is mainly applied to speculators, then these alpha portfolios are suddenly exposed to huge systematic risk. Given the fact that institutional investors are ordered by the CSRC not to sell their shares in our sample period, futures

⁵The basis calculated as $\log(\text{futures Price}) - \log(\text{Spot Price})$ are 3.30% and 4.5% in our sample period for the nearest maturity contracts of CSI 300 and CSI 500 respectively. See Figure 3.

trading restrictions imply that most likely these institutional investors would choose not to buy new shares of component stocks.

In a summary, index arbitragers and hedgers can not be active traders in the spot market because of the persistent negative basis during our sample. They are not the reason for our findings as the negative basis existed long before the two rounds of futures trading restrictions. Instead, we believe that loss of trading incentives of “alpha-traders” is the main reason for the deterioration of spot market quality.

5 Conclusions

We use the recent Chinese stock market crash and the subsequent index futures trading restrictions as a quasi-natural experiment to examine the effect of index futures trading on spot market quality. A difference-in-difference approach is adopted to identify the net effect. We find that after the Chinese futures exchange effectively terminates index futures trading, the spot market quality is significantly lowered. That is, the index component stocks quality is affected more by the termination of futures trading than is the quality of non-component stocks. We also find that the impact is stronger for the September round of trade restrictions for the CSI 300 index. Our results cannot be explained by the informed trading theory, and are inconsistent with the index arbitrage story. The changing trading behavior of “alpha-traders” after futures trading restrictions is the main reason for the deterioration of market quality.

With that, however, we do recognize that more detailed information such as investor trading records and limit order books are needed to further confirm our interpretations. Further studies are warranted when data is available.

Our results may have some important policy implications. Since the market crash, Chinese regulators and the public have cast serious doubts on the functioning of index

futures. So far there has been no sign that index futures trading will be resumed any time soon. We provide a rigorous analysis on the effect of the index futures trading restriction. As long as the index futures trading is not resumed, the resulting deterioration of spot market quality will continue to distort the market price and lead to capital misallocation.

In addition, after the CFFEX trade restriction regulations, the trading volume of Singapore A50 index futures soared as international investors rushed to an alternative to hedge their risk exposure to the Chinese A share market. On October 12, 2015, the Chicago Mercantile Exchange (CME) launched its FT China 50 index futures. It is plausible that if Chinese regulators keep the door of futures trading shut, then the CFFEX may lose ground to its overseas competitors.

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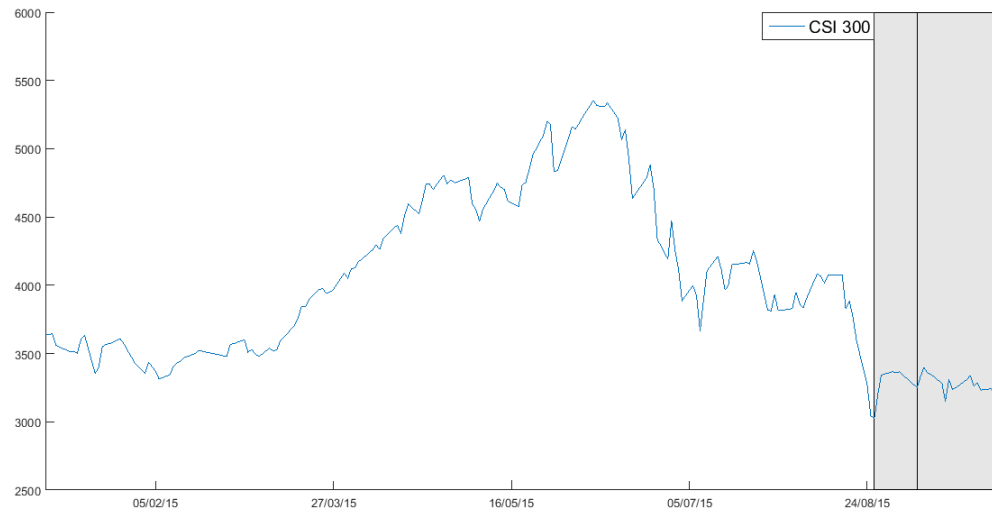
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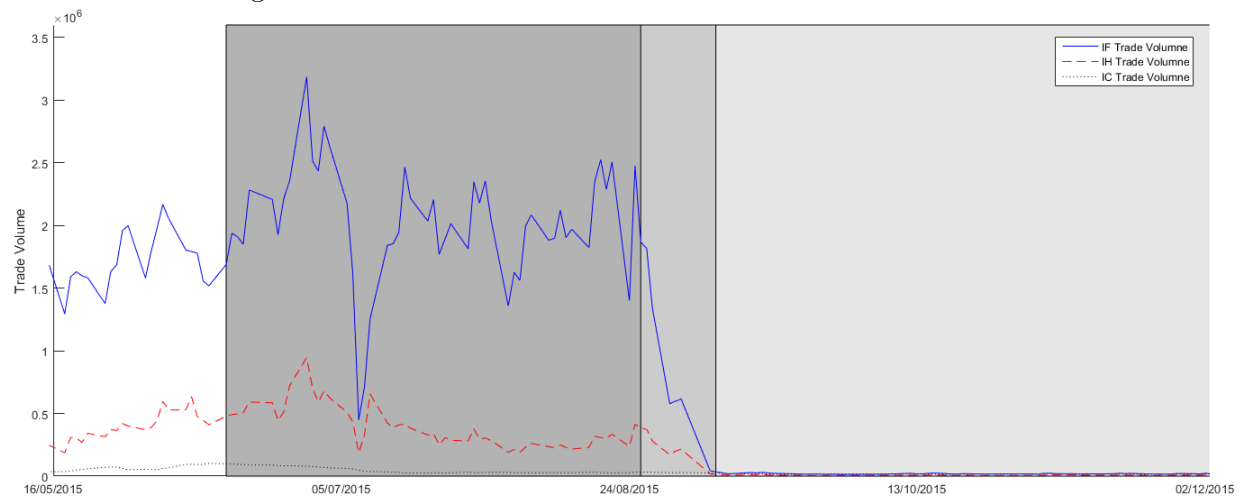
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Figure 1: The Recent Chinese Stock Market Crash



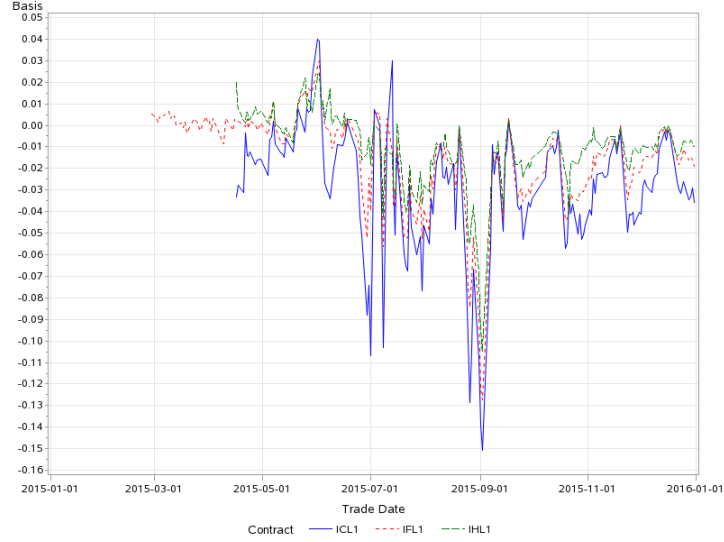
This figure shows the time series of the CSI 300 index from Apr 16, 2015 to September 30, 2015. The two greyed regions correspond to the two event windows as defined in the main text: Aug. 26, 2015 to Sept. 06, 2015 and Sept. 07, 2015 to Sept. 30, 2015.

Figure 2: Total trade Volume of Index Futures Contract



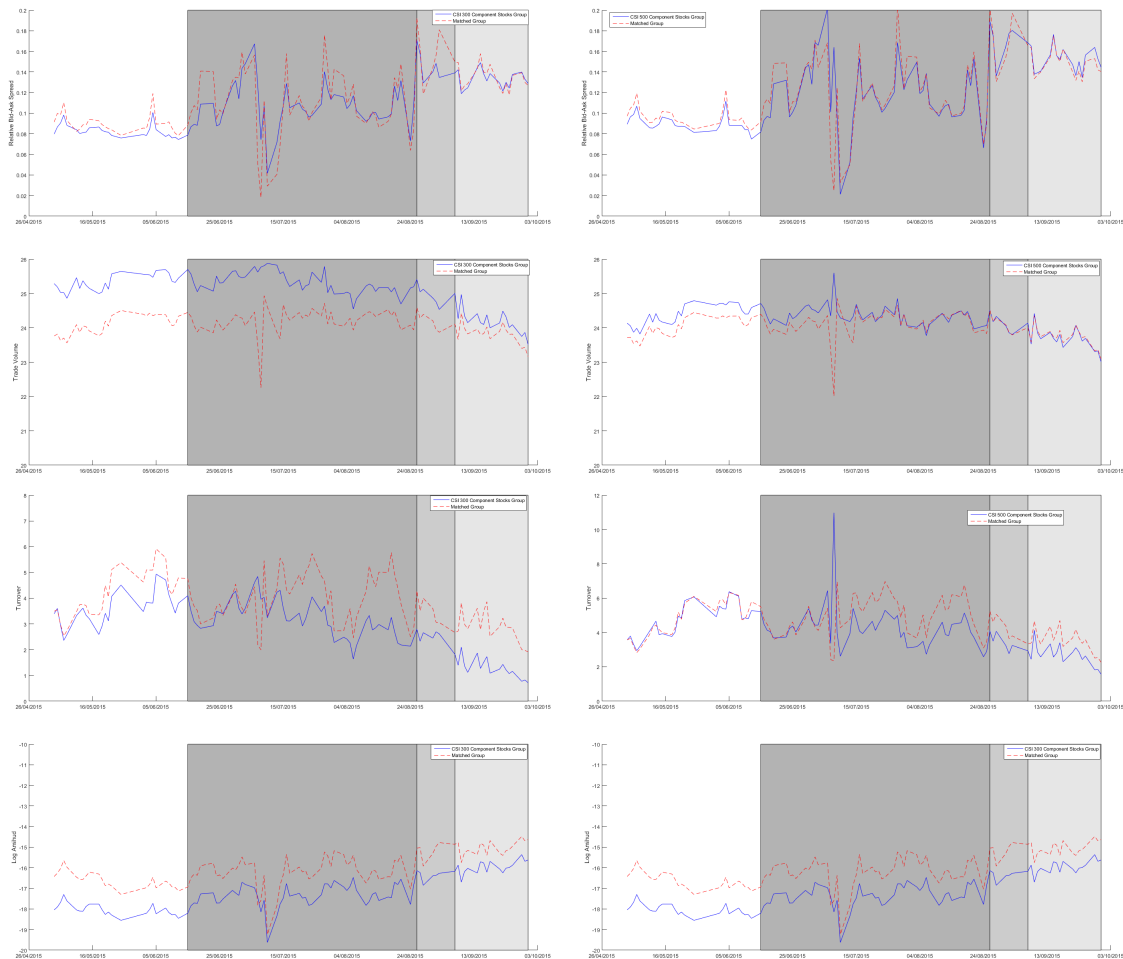
This figure shows the time series of trade volume of three index futures contract from May 16, 2015 to September 31, 2015. The first greyed region corresponds to Jun. 15 - Aug. 26 2015. The second and third greyed regions correspond to the two event windows as defined in the main text: Aug. 26, 2015 to Sept. 06, 2015 and Sept. 07, 2015 to Dec. 31, 2015 respectively.

Figure 3: Basis on Prices Three the Nearest-Maturity Index Futures Contract



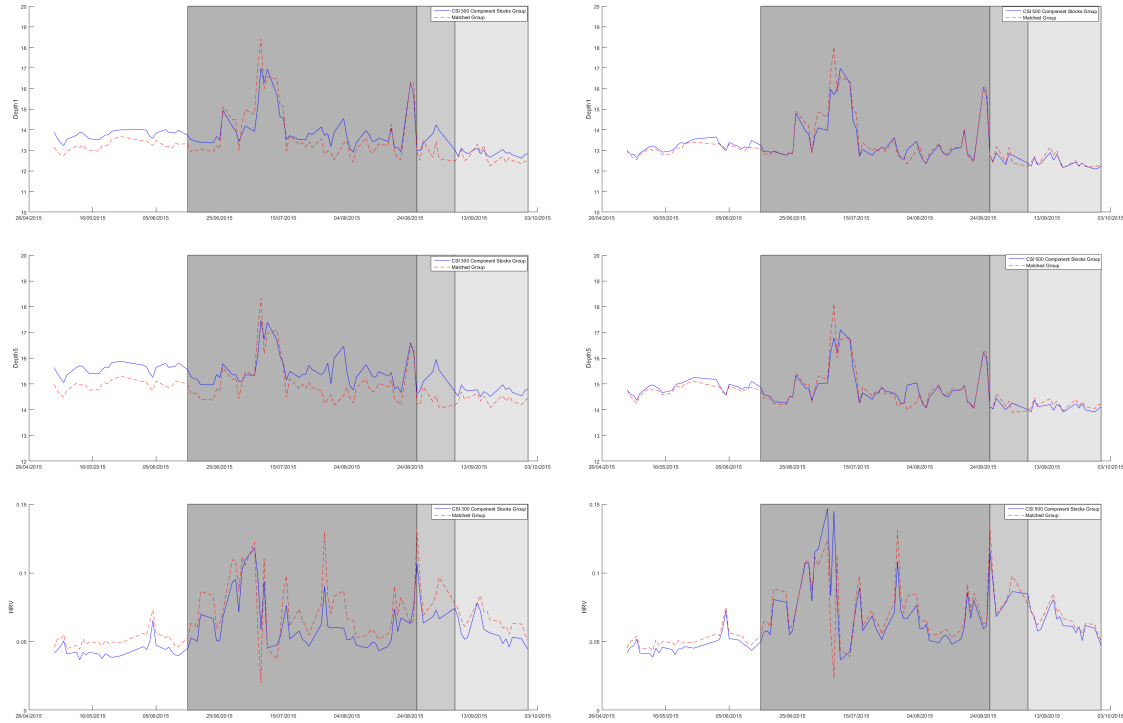
This figure shows the time series of basis of IC, IF and IH nearest-maturity index futures contracts from May 16, 2015 to September 31, 2015. Each basis is calculated as $\log(F_t) - \log(S_t)$, where F_t and S_t are futures and spot prices, respectively. The first greyed region corresponds to Jun. 15 - Aug. 26 2015. The second and third greyed regions correspond to the two event windows as defined in the main text: Aug. 26, 2015 to Sept. 06, 2015 and Dec. 07, 2015 to Sept. 31, 2015 respectively.

Figure 4: Average Market Quality Series of the CSI 300 and CSI 500 Component Stocks and Non-Component stocks



This figure shows the average relative bid-ask spread, $\log(\text{tradeVolume})$, $\log(\text{turnover})$ and $\log(\text{amihud})$ time series of index component and non-component stocks from Apr. 16, 2015 to Sept 30, 2015. The left column corresponds to the series of the CSI 300 index case and the right column corresponds to the CSI 500 index case. The three greyed regions correspond to the periods of Jun. 15th, 2015 to Aug. 26th, 2015, Aug. 26th, 2015 to Sept. 07th, 2015, Sept. 07th, 2015 to Sept. 30th, 2015, respectively.

Figure 5: Average Market Quality Series of the CSI 300 and the CSI 500 Component Stocks and Non-Component Stocks



This figure shows the average $\log(\text{Depth1})$, $\log(\text{Depth5})$ and $\log(\text{HFV})$ time series of index component and non-component stocks from Apr. 16, 2015 to Sept 30, 2015. The left column corresponds to the series of the CSI 300 index case and the right column corresponds to the CSI 500 index case. The three greyed regions correspond to the periods of Jun. 15th, 2015 to Aug. 26th, 2015; Aug. 26th, 2015 to Sept. 07th, 2015; and Sept. 07th, 2015 to Sept. 30th, 2015, respectively.

Table 1: Summary Statistics

Sample Size	All stocks		CSI 300 Component		CSI 500 Component		Non-Component	
	1037		161		264		612	
	Mean	Std	Mean	Std	Mean	Std	Mean	Std
<i>RPD</i>	0.13	0.06	0.11	0.05	0.12	0.05	0.14	0.06
<i>Log(Trade Volume)</i>	24.14	1.22	25.11	1.25	24.26	1.09	23.81	1.10
<i>Turnover</i>	4.70	3.66	2.94	2.41	4.08	2.92	5.47	4.00
<i>Log(Amihud)</i>	-15.96	1.45	-17.22	1.24	-16.16	1.29	-15.52	1.35
<i>Log(Depth1)</i>	13.03	1.38	13.67	1.33	13.10	1.34	12.82	1.35
<i>Log(Depth5)</i>	14.55	1.05	15.35	1.14	14.63	0.96	14.29	0.94
<i>HRV</i>	0.13	0.08	0.11	0.08	0.13	0.08	0.14	0.08
<i>Bsratio</i>	0.46	0.07	0.47	0.06	0.46	0.07	0.46	0.07
<i>RV</i>	5.44	3.07	4.65	2.70	5.27	2.90	5.74	3.18
<i>Log(Fsize)</i>	16.24	0.97	17.79	0.76	16.45	0.48	15.71	0.62
<i>Invp</i>	0.08	0.05	0.08	0.05	0.08	0.04	0.08	0.05
<i>Log(Trade Money)</i>	26.83	1.13	27.86	1.08	27.00	1.00	26.46	0.99

This table presents the daily summary statistics for the whole sample period from May 04, 2015 to Sep. 30, 2015. *RPD*, *Log(Trade Volume)*, *Turnover* *Log(Amihud)*, *Log(Depth1)*, *Log(Depth5)*, *HRV*, *Log(Fsize)* and *Invp* are relative bid-ask spread, log trade volume, turnover, log *amihud*, log market depth1, log market depth5, high frequency volatility, log firm size and inverse of price respectively. The second, forth, sixth, eighth columns correspond means of row variables of all of sample, CSI 300 component, CSI 500 component and non-component stocks respectively. The third, fifth, seventh and ninth and are standard deviations of row variables of all of sample, CSI 300 component, CSI 500 component and non-component stocks respectively. *RV* are computed using intraday 1 minute stock price data for each firm *i*. *Log(Fsize)* is the logarithm of an individual stock's market value. All data are from CSMAR.

Table 2: Difference-in-Difference Analysis for Relative Bid-Ask Spread

	CSI 300		CSI 500	
	2015/8/26	2015/9/7	2015/8/26	2015/9/7
Intercept	0.0195 (0.0299)	0.0145 (0.0301)	0.0365 (0.0240)	0.0368 (0.0242)
$RV_{i,t}$	0.0041*** (0.0005)	0.0041*** (0.0005)	0.0043*** (0.0004)	0.0043*** (0.0004)
$Invp_{i,t}$	0.7061*** (0.0402)	0.7071*** (0.0400)	0.7081*** (0.0377)	0.7079*** (0.0376)
$Log(Trade\ Amount_{i,t})$	0.0030** (0.0012)	0.0032*** (0.0012)	0.0028*** (0.0010)	0.0028*** (0.0010)
D_i	-0.0927*** (0.0060)	-0.0940*** (0.0060)	-0.0785*** (0.0014)	-0.0783*** (0.0014)
D_t	0.0235*** (0.0034)	0.0223*** (0.0033)	0.0254*** (0.0027)	0.0251*** (0.0027)
$D_i \times D_t$	0.0008 (0.0021)	0.0035* (0.0020)	0.0040** (0.0019)	0.0046*** (0.0017)
Adj. R-Squared	0.720	0.720	0.660	0.660

where $RV_{i,t}$, $Invp_{i,t}$, and $Log(Trade\ Amount_{i,t})$ are the 1-minute realized volatility, inverse of price and log trade amount for firm i at t respectively. $D_i = 1$ if i is a component stock of the CSI index, $D_i = 0$ otherwise; $D_t = 1$, if $t \geq Event\ date$, $D_t = 0$ otherwise. The values in the brackets are t statistics calculated by using robust standard errors clustered at the stock level for testing $H_0 : |\beta_i| = 0$ ($H_1 : |\beta_i| \neq 0$) for $i \in \{0, 1, \dots, 7\}$. *, ** and *** denote significance at 10%, 5% and 1% respectively.

Table 3: Difference-in-Difference Analysis for Trade Volume

	CSI 300		CSI 500	
	2015/8/26	2015/9/7	2015/8/26	2015/9/7
Intercept	15.9210*** (2.3776)	15.9272*** (2.3556)	12.2777*** (2.3998)	12.2724*** (2.3918)
$RV_{i,t}$	0.0805*** (0.0084)	0.0818*** (0.0085)	0.0911*** (0.0070)	0.0913*** (0.0070)
$\text{Log}(Fsize_{i,t})$	0.3436** (0.1416)	0.3439** (0.1403)	0.5558*** (0.1429)	0.5566*** (0.1424)
invp	0.8278*** (0.1323)	0.6713*** (0.1299)	0.8123*** (0.1482)	0.7927*** (0.1479)
D_i	4.2710*** (0.2847)	4.2497*** (0.2830)	3.5980*** (0.0247)	3.5806*** (0.0231)
D_t	-0.7954*** (0.0918)	-0.7591*** (0.0916)	-0.6077*** (0.0766)	-0.6101*** (0.0763)
$D_i \times D_t$	-0.5698*** (0.0590)	-0.6409*** (0.0620)	-0.2513*** (0.0441)	-0.2461*** (0.0467)
Adj. R-Squared	0.848	0.849	0.780	0.779

where $RV_{i,t}$, $\log(Fsize_{i,t})$, and $Return_{i,t}$ are the 1-minute realized volatility, log firm size and daily return for firm i at t respectively. $D_i = 1$ if i is a component stock of the CSI index, $D_i = 0$ otherwise; $D_t = 1$, if $t \geq \text{Event date}$, $D_t = 0$ otherwise. The values in the brackets are t statistics calculated by using robust standard errors clustered at the stock level for testing $H_0 : |\beta_i| = 0$ ($H_1 : |\beta_i| \neq 0$) for $i \in \{0, 1, \dots, 7\}$. *, ** and *** denote significance at 10%, 5% and 1% respectively.

Table 4: Difference-in-Difference Analysis for Turnover

	CSI 300		CSI 500	
	2015/8/26	2015/9/7	2015/8/26	2015/9/7
Intercept	-43.5822*** (7.6601)	-43.4689*** (7.6409)	-40.4865*** (10.1365)	-40.5882*** (10.1441)
$RV_{i,t}$	0.2679*** (0.0311)	0.2691*** (0.0312)	0.3346*** (0.0299)	0.3347*** (0.0299)
$\text{Log}(Size_{i,t})$	3.5994*** (0.4856)	3.5923*** (0.4845)	3.0052*** (0.6853)	3.0130*** (0.6858)
D_i	-22.0736*** (1.5202)	-22.0589*** (1.5207)	-7.0317*** (1.3256)	-7.0747*** (1.3268)
D_t	-0.3363 (0.3107)	-0.2625 (0.3075)	-0.7941*** (0.2752)	-0.8263*** (0.2732)
$D_i \times D_t$	-0.6115*** (0.1537)	-0.7597*** (0.1571)	-0.2914* (0.1590)	-0.2240 (0.1569)
Adj. R-Squared	0.621	0.621	0.548	0.547

where $RV_{i,t}$, $\log(Size_{i,t})$ are the 1-minute realized volatility, log firm size for firm i at t respectively. $D_i = 1$ if i is a component stock of the CSI index, $D_i = 0$ otherwise; $D_t = 1$, if $t \geq \text{Event date}$, $D_t = 0$ otherwise. The values in the brackets are t statistics calculated by using robust standard errors clustered at the stock level for testing $H_0 : |\beta| = 0$ ($H_1 : |\beta| \neq 0$). *, ** and *** denote significance at 10%, 5% and 1% respectively.

Table 5: Difference-in-Difference Analysis for Log Amihud

	CSI 300		CSI 500	
	2015/8/26	2015/9/7	2015/8/26	2015/9/7
Intercept	-3.9829 (4.1297)	-4.0114 (4.1068)	8.0936** (3.6783)	8.1094** (3.6774)
$RV_{i,t}$	0.0718*** (0.0046)	0.0711*** (0.0045)	0.0681*** (0.0031)	0.0680*** (0.0031)
$\text{Log}(Size_{i,t})$	-0.7129*** (0.2451)	-0.7114*** (0.2438)	-1.4188*** (0.2186)	-1.4201*** (0.2185)
D_i	-1.2367** (0.4878)	-1.2332** (0.4869)	-1.6690*** (0.0334)	-1.6555*** (0.0313)
D_t	1.7641*** (0.1406)	1.7384*** (0.1390)	1.3210*** (0.1073)	1.3262*** (0.1059)
$D_i \times D_t$	0.2693*** (0.0688)	0.3187*** (0.0696)	0.1746*** (0.0511)	0.1635*** (0.0523)
Adj. R-Squared	0.802	0.802	0.731	0.731

where $RV_{i,t}$, $\log(Size_{i,t})$ are the 1-minute realized volatility and log firm size for firm i at t respectively. $D_i = 1$ if i is a component stock of the CSI index, $D_i = 0$ otherwise; $D_t = 1$, if $t \geq \text{Event date}$, $D_t = 0$ otherwise. The values in the brackets are t statistics calculated by using robust standard errors clustered at the stock level for testing $H_0 : |\beta| = 0$ ($H_1 : |\beta| \neq 0$). *, ** and *** denote significance at 10%, 5% and 1% respectively.

Table 6: Difference-in-Difference Analysis for *Depth1*

	CSI 300		CSI 500	
	2015/8/26	2015/9/7	2015/8/26	2015/9/7
Intercept	2.5400 (4.0997)	2.4515 (4.1724)	-2.3631 (4.4857)	-2.3527 (4.4741)
$RV_{i,t}$	-0.0701*** (0.0075)	-0.0694*** (0.0074)	-0.0640*** (0.0055)	-0.0640*** (0.0055)
$\log(Fsize_{i,t})$	0.7465*** (0.2706)	0.7660*** (0.2765)	1.1533*** (0.3115)	1.1516*** (0.3104)
$\log(Trade\ Amount_{i,t})$	-0.0711 (0.0551)	-0.0804 (0.0557)	-0.1366*** (0.0431)	-0.1358*** (0.0431)
D_i	-0.0172 (0.5664)	-0.0130 (0.5791)	0.4637*** (0.1346)	0.4569*** (0.1355)
D_t	-0.6685*** (0.1622)	-0.6156*** (0.1616)	-0.4622*** (0.1362)	-0.4640*** (0.1349)
$D_i \times D_t$	-0.0714 (0.0713)	-0.1919** (0.0752)	-0.0885** (0.0440)	-0.0839* (0.0431)
Adj. R-Squared	0.510	0.510	0.522	0.522

where $RV_{i,t}$, $\log(Fsize_{i,t})$, and $\log(Trade\ Amount_{i,t})$ are the 1-minute realized volatility, log firm size and log trade amount for firm i at t respectively. $D_i = 1$ if i is a component stock of the CSI index, $D_i = 0$ otherwise; $D_t = 1$, if $t \geq Event\ date$, $D_t = 0$ otherwise. The values in the brackets are t statistics calculated by using robust standard errors clustered at the stock level for testing $H_0 : |\beta| = 0$ ($H_1 : |\beta| \neq 0$). *, ** and *** denote significance at 10%, 5% and 1% respectively.

Table 7: Difference-in-Difference Analysis for *Depth5*

	CSI 300		CSI 500	
	2015/8/26	2015/9/7	2015/8/26	2015/9/7
Intercept	8.5281*** (2.9482)	8.4568*** (3.0446)	3.1879 (3.3868)	3.2023 (3.3756)
$RV_{i,t}$	-0.0604*** (0.0067)	-0.0595*** (0.0066)	-0.0523*** (0.0055)	-0.0522*** (0.0044)
$\text{Log}(\text{Size}_{i,t})$	0.5248*** (0.2009)	0.5479*** (0.2089)	1.1533*** (0.3115)	0.8417*** (0.2212)
$\text{Log}(\text{Trade Amount}_{i,t})$	-0.0820* (0.0424)	-0.0934** (0.0429)	-0.1366*** (0.0431)	-0.1291*** (0.0352)
D_i	0.7263 (0.5156)	0.7097 (0.5346)	0.4637*** (0.1346)	1.6659*** (0.0960)
D_t	-0.5901*** (0.1213)	-0.5200*** (0.1218)	-0.4622*** (0.1362)	-0.4126*** (0.1029)
$D_i \times D_t$	-0.1293** (0.0570)	-0.2863*** (0.0606)	-0.0885** (0.0440)	-0.1550*** (0.0345)
Adj. R-Squared	0.594	0.595	0.540	0.540

where $RV_{i,t}$, $\log(\text{Size}_{i,t})$, and $\text{Log}(\text{Trade Amount}_{i,t})$ are the 1-minute realized volatility, log firm size and log trade money for firm i at t respectively. $D_i = 1$ if i is a component stock of the CSI index, $D_i = 0$ otherwise; $D_t = 1$, if $t \geq \text{Event date}$, $D_t = 0$ otherwise. The values in the brackets are t statistics calculated by using robust standard errors clustered at the stock level for testing $H_0 : |\beta| = 0$ ($H_1 : |\beta| \neq 0$). *, ** and *** denote significance at 10%, 5% and 1% respectively.

Table 8: Difference-in-Difference Analysis for High Frequency Volatility

	CSI 300		CSI 500	
	2015/8/26	2015/9/7	2015/8/26	2015/9/7
Intercept	-0.3750*** (0.0388)	-0.3774*** (0.0386)	-0.4084*** (0.0426)	-0.4099*** (0.0426)
$\text{Log}(\text{Trade Amount}_{i,t})$	0.0128*** (0.0005)	0.0130*** (0.0005)	0.0911*** (0.0070)	0.0152*** (0.0004)
$\text{Log}(\text{Fsize}_{i,t})$	0.0052** (0.0024)	0.0051** (0.0024)	0.5558*** (0.1429)	0.0034 (0.0026)
$\text{Inv}p_{i,t}$	0.2570*** (0.0207)	0.2602*** (0.0206)	0.8123*** (0.1482)	0.2646*** (0.0206)
D_i	-0.0790*** (0.0053)	-0.0798*** (0.0053)	3.5980*** (0.0247)	-0.0361*** (0.0017)
D_t	0.0155*** (0.0015)	0.0141*** (0.0015)	-0.6077*** (0.0766)	0.0159*** (0.0011)
$D_i \times D_t$	0.0049*** (0.0010)	0.0079*** (0.0011)	-0.2513*** (0.0441)	0.0037*** (0.0008)
Adj. R-Squared	0.630	0.616	0.616	0.616

where $\text{Log}(\text{Trade Amount}_{i,t})$, $\text{log}(\text{Fsize}_{i,t})$ and $\text{Inv}p_{i,t}$ are the log daily trade amount, log firm size and inverse of stock price for firm i at t respectively. $D_i = 1$ if i is a component stock of the CSI index, $D_i = 0$ otherwise; $D_t = 1$, if $t \geq \text{Event date}$, $D_t = 0$ otherwise. The values in the brackets are t statistics calculated by using robust standard errors clustered at the stock level for testing $H_0 : |\beta_i| = 0$ ($H_1 : |\beta_i| \neq 0$) for $i \in \{0, 1, \dots, 7\}$. *, ** and *** denote significance at 10%, 5% and 1% respectively.